Mahdi Asefi

AI Engineer — LLM Systems, Computer Vision & Generative AI

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Open to new roles in AI engineering — LLM & agent systems, computer vision, and generative AI.

SUMMARY

AI Engineer with 4+ years building production AI systems across LLM agents, computer vision, and generative AI. Shipped 3 AI products and 7+ models to production at PiRun (Drivee.ai); now sole engineer on an end-to-end generative video platform at Metadeep. M.Sc. in Artificial Intelligence (GPA 18.89/20), thesis in progress on face forgery detection.

TECHNICAL SKILLS

Languages & Core: Python · SQL · Git · Linux · Docker

Deep Learning & Vision: PyTorch · Hugging Face Transformers · T5 / mT5 · scikit-learn · TensorFlow · OpenCV · YOLO · ViT · CNNs · GANs · Matplotlib

LLM & Agent Systems: OpenAI API · LangChain · LangGraph · RAG · Qdrant · FAISS · prompt engineering · tool & function calling · multi-agent systems · LLM evaluation & fine-tuning · Langfuse · Redis-backed conversational memory · ElevenLabs (TTS / voice cloning)

Generative Media & Video: ComfyUI · ComfyUI API · ControlNet · IP-Adapter · VAE · diffusion models · Vision LLM / VLM · FFmpeg · Vast.ai (on-demand cloud GPU) · image & video prompt engineering

MLOps & Infrastructure: model deployment & serving · ONNX · MLflow · CI/CD · async & background job queues

Domain & Research: deepfake & face forgery detection · face detection & identification · GANs for medical imaging · synthetic data generation · Persian NLP · question answering systems · multimodal / VLM pipelines · image processing · data cleaning & augmentation · hyperparameter tuning

EXPERIENCE

Full-Stack AI Engineer — Metadeep
Karaj, Alborz, Iran · Hybrid · Feb 2026 – Present

- Architected an end-to-end generative video platform from scratch as sole engineer, owning frontend, backend, AI pipeline, and generation infrastructure.
- Designed a multi-agent orchestration system in LangGraph for automated video scripting, with human-in-the-loop verification checkpoints across the pipeline.
- Integrated and benchmarked 5+ generative video and image models through the ComfyUI API (ControlNet, IP-Adapter, VAE), comparing open-source and commercial generators.
- Built a Vision LLM asset-analysis engine that extracts visual properties and auto-generates structured creative briefs, compiled into shot-level prompts across cinematic styles.
- Architected on-demand GPU provisioning on Vast.ai, spinning up compute per job rather than keeping dedicated instances idle.
- Integrated a Persian/Farsi voice-over pipeline (voice cloning and TTS) with RTL subtitle rendering.

AI Engineer — Tetronix
Remote (Canada) · Sep 2025 – Dec 2025

- Built and deployed an AI chatbot on FastAPI using the OpenAI API, with Langfuse for prompt management and observability.
- Implemented short-term conversational memory in Redis and long-term memory via message summarisation, cutting context/token usage by ~30%.
- Integrated real-time voice interaction with ElevenLabs, sharing conversational state with the FastAPI backend through Redis.
- Developed a scalable web-crawling pipeline aggregating car-marketplace data into MongoDB, with similarity-based vehicle retrieval for related-car search.

AI Engineer — PiRun (Drivee.ai)
Remote (US) · Mar 2023 – Jul 2025

- Joined early and contributed to three AI products shipped in production: Drivee AI, Drivee Sales Agent, and HeyGent.
- Took 7+ models to production — foreground segmentation, car-part detection (YOLO), lighting enhancement, depth-based bokeh, and CGAN shadow and reflection generation.
- Built multi-agent systems for sales, appointment booking, and customer engagement, handling hundreds of leads and bookings per month via structured outputs and tool calling.
- Applied CGANs to generate realistic shadows and reflections for virtual environments, optimising models for fast CPU inference through ONNX.
- Evaluated and fine-tuned LLMs, authored modular prompts, and integrated chatbot systems with CRM, DMS, and scheduling tools.

NLP Engineer Intern — Commercialization and Innovation Center, University of Kashan
Isfahan, Iran · Apr – Jun 2022

- Fine-tuned a multilingual (mT5) question answering system on the QA task using Hugging Face Transformers.

Computer Vision Engineer (Part-time) — Sina Teb
Isfahan, Iran · Mar – Oct 2021

- Fine-tuned a YOLOv5 model on open-source datasets to build a car detection system for long-distance detection in UAV aerial imagery, then developed and deployed a real-time car detection and tracking system running on server.

Teaching Assistant — Fundamentals of Computer and Programming, University of Kashan
Kashan, Isfahan, Iran · Spring 2019

- Designed course assignments and taught programming fundamentals (C++) with hands-on practice in Visual Studio, guiding approximately 10 students through debugging sessions and grading their work.

RESEARCH EXPERIENCE

Deepfake and Face Forgery Detection — Research Assistant
Supervisor: Prof. Azadeh Mansouri (Kharazmi University) · 2024 – Present

- Reviewed and synthesised five deepfake-detection papers, presenting findings and methodologies to faculty and the research team.
- Trained models from two of the reviewed papers, evaluated their performance and limitations, and developed a combined approach for improved face deepfake detection.
- Thesis project for the M.Sc. in Artificial Intelligence; the RFAM (AAAI 2021) implementation and its combination with CORE (CVPRW 2022) are published as open-source code.

GANs in Radiogenomics — Research Assistant
Supervisor: Prof. Qian Liu · Winter 2023

- Studied GAN-based synthetic data generation for breast cancer detection, combining MRI imagery with genomics information.
- Investigated fusing genomic data with MRI feature maps into a unified generation network, and presented papers on GAN-based augmentation for radiogenomics models.

Face Detection and Clustering in News Imagery — Research Assistant
Supervisor: Dr. Pooria Lajevardy · March 2022

- Summarised and analysed two papers on face detection and identification in news images, critiquing their methodologies and proposing alternative approaches to the research team.

EDUCATION

M.Sc. in Artificial Intelligence — Azad University of Tehran Central Branch
Tehran, Iran · 2023 – Present (Expected 2026) · GPA 18.89/20

Thesis: Deepfake and face forgery detection. Supervisor: Prof. Reyhaneh Tati (Azad University) · Consulting Supervisor: Prof. Azadeh Mansouri (Kharazmi University)

B.Sc. in Computer Science — University of Kashan
Isfahan, Iran · 2018 – Jan 2023 · GPA 16.64/20

Awarded a stipend for exceptional performance on the undergraduate research project "Persian Paraphrasing". Supervisor: Prof. Mahsa Soheil Shamaee

SELECTED PROJECTS

Combined Face Forgery Detection Method (RFAM × CORE)
github.com/mahdiAsefi20/RFAMxCORE

- Implemented a unified training pipeline coupling RFAM's frequency-aware cues with CORE's consistency objective on an Xception backbone.
- Trained and evaluated on FaceForensics++, with a dedicated inference path for the FF++ low-quality manipulation splits.

RFAM — Local Relation Learning for Face Forgery Detection (AAAI 2021)
github.com/mahdiAsefi20/RFAM

- Unofficial from-scratch reimplementation of the paper's RGB-frequency attention, frequency-aware cue, and multi-scale patch similarity modules.
- Built the full dataset pipeline for FaceForensics++, DFDC, DFFD, and Celeb-DF.

BERT from Scratch
github.com/mahdiAsefi20/miniBERT

- Implemented a complete BERT in PyTorch — tokenizer, embedding layer, encoder stack, and training loops — then pretrained it via masked language modelling on WikiText and news corpora, with a classification fine-tuning pipeline and evaluation metrics.

Persian Paraphrasing
Undergraduate Research Project, University of Kashan · Jul 2022 – Nov 2022 · github.com/mahdiAsefi20/Persian-Paraphraser

- Translated and curated a Persian paraphrase corpus from the 50M-pair ParaNMT English–Persian dataset, chunking translation to fit Colab memory limits and filtering low-similarity pairs via Levenshtein distance.
- Fine-tuned mT5 for Persian paraphrase generation and co-authored a 14-page paper with Prof. Mahsa Soheil Shamaee.

LANGUAGES

Persian — Native

English — Professional working proficiency (TOEFL iBT 91/120: Reading 24 · Listening 24 · Speaking 20 · Writing 23)